Up to which $k_{\max}$ is linear galaxy bias enough?

A linear-bias fit to a one-loop truth on four bin layouts

November 2026

1 Summary

The question. The forecast fits the DESI galaxy clustering band powers C_ℓ^{gg} of thin redshift bins with linear galaxy bias: the galaxy density contrast is $\delta_g = b_1\,\delta$, where δ is the matter density contrast and b_1 the linear bias of a bin. This is exact on large scales. On smaller scales higher-order terms grow with the wavenumber k, so linear bias is safe only below some Fourier scale cut $k_{\max}$. This note answers two questions about the correlated-prior note (`notes/correlated_bias_prior`, "the v1 note" below).

1. Up to which $k_{\max}$ does a linear-bias fit to a one-loop truth stay unbiased, on all four bin layouts, for the dark-energy pair (w_0, w_a), for Ω_m and for A_s?
2. Does the v1 accuracy test of the bias priors, tied versus correlated, give the same verdict for Ω_m and A_s as for (w_0, w_a)?

Here $w(a) = w_0 + w_a(1-a)$ is the dark-energy equation of state, with a the scale factor; Ω_m is the matter density today in units of the critical density; and A_s is the amplitude of the primordial curvature power spectrum, sampled as $10^9 A_s$.

The short answer. No scanned cut is safe on all four layouts. Only the layout of equal comoving bin width passes, at $k_{\max} = 0.05\,h/\mathrm{Mpc}$, on every rung of the bias ladder and for every truth tried. At $k_{\max} = 0.075\,h/\mathrm{Mpc}$, the cut of the v1 note, a linear fit is biased on every layout and every rung, and freeing the linear bias bin by bin does not cure it. The safe cut is also expensive: it keeps less than a tenth of the dark-energy figure of merit, the inverse area of the (w_0, w_a) error ellipse. For the second question the answer is yes: scored for Ω_m and A_s, the v1 test excludes the tied model and accepts the correlated priors, as it does for (w_0, w_a).

The setup. The data are the DESI band powers C_ℓ^{gg} alone, computed without the Limber approximation, for the six tracer groups BGS, LRG1–3 and ELG1–2, on four bin layouts: bins of equal comoving width $\Delta d_{\mathrm{com}} = 62$ Mpc, and bins of fixed redshift width $\Delta z = 0.02$, 0.05 and 0.2. The Gaussian covariance is computed for 14,000 square degrees and rescaled to the survey's 12,300. Each one-loop truth, described below, is fitted with linear bias on the four rungs of the v1 ladder. With $D(z)$ the linear growth factor normalized to $D(0) = 1$ and z_i the redshift of bin i, they are: **tied**, one free amplitude B_g per group g and $b_i = B_g/D(z_i)$; **baseline**, a free amplitude and tilt per group plus a correlated residual prior of width $\sigma_g = 3\%$ for BGS and 1% for the other groups, with correlation length $L = 0.2$ in redshift; **pessimistic**, the same with $\sigma_g = 3\%$ everywhere and $L = 0.05$; **independent**, one free bias per bin. The error of each fit is its Fisher linear response (Appendix A). The (w_0, w_a) shift is scored in

units of the rung's own marginalized error ellipse, and Ω_m and A_s each in units of its own marginalized error σ. A fit is unbiased when every shift is below $\sigma/3$, the v1 note's threshold.

The scale cut. $k_{\rm max}$ labels a family of cuts in multipole, one per bin. Bin i is cut at $k_{\rm max}/D(\bar{z}_i)$, with $\bar{z}_i$ its mean redshift: the density field is less nonlinear at higher redshift, so the cut moves to higher k as $1/D$. The cut is applied through the non-Limber ultraviolet-fraction criterion of Saraivanov et al. [1], which follows DeRose and Chen [2]. Each band power is split into $C_\ell^{\rm IR}$, computed from the power spectrum below the bin's cut wavenumber, and $C_\ell^{\rm UV}$, from the power above it. A band is kept while

$$\frac{|C_\ell^{\rm UV}|}{|C_\ell^{\rm UV}| + |C_\ell^{\rm IR}|} \;<\; \alpha = 0.05, \tag{1}$$

that is, while less than 5% of the band comes from beyond the cut. The kept bands are intersected with the shipped mask, which caps every bin at $k = 0.1185\,{\rm Mpc}^{-1}$ with no growth scaling. The scan runs over $k_{\rm max} = 0.05, 0.075, \ldots, 0.25\,h/{\rm Mpc}$, and every $k_{\rm max}$ in this note is this $z = 0$ value.

The truths. The truth is model A of `notes/higher_order_bias`, computed with CosmoLike's one-loop engine. Its real-space galaxy power spectrum is

$$P_{gg}(k) \;=\; b_1^2\,P_{\rm mm}(k) \;+\; P_{\rm loop}(k;\, b_1, b_2, b_{s^2}) \;+\; \alpha_0\,k^2 P_{\rm mm}(k), \tag{2}$$

where $P_{\rm mm}$ is the nonlinear matter power spectrum and $P_{\rm loop}$ the one-loop terms. These carry the quadratic bias b_2, the response of the galaxy density to δ^2, and the tidal bias b_{s^2}, its response to the squared tidal field. α_0 is the k^2 counterterm, in $h^{-2}\,{\rm Mpc}^2$, which absorbs the small-scale sensitivity of the loops. Model A's definition, like DESI's, multiplies the linear spectrum $P_{\rm lin}$, while CosmoLike's counterterm slot, $2\,b_1 b_K\,k^2 P_{\rm mm}$ with b_K its per-bin derivative-bias coefficient and $\alpha_0 = 2\,b_1 b_K\,(c/H_0)^2$, multiplies the nonlinear one; the truths use CosmoLike's convention (subsection "The draft's data vector against CosmoLike" of `notes/higher_order_bias`). The *calibrated* truth keeps the fiducial b_1 and puts b_2 on the Lazeyras et al. fit $b_2(b_1)$ [3] and $b_{s^2} = -\frac{4}{7}(b_1 - 1)$ on co-evolution [4], with third-order bias $b_3 = 0$, $\alpha_0 = 0$ and no stochastic term. Four brackets offset every chunk's b_2 by ± 1 ("$b_2 \pm 1$", model A's prior width) or set $\alpha_0 = \pm 12.5\,h^{-2}\,{\rm Mpc}^2$ (the counterterm prior width of the DESI DR1 full-shape analysis [5]). "Every truth" means all five.

The safe cut, layout by layout. Table 1 lists the largest safe $k_{\rm max}$: the largest cut at which every shift stays below $\sigma/3$, at that cut and at every smaller cut that has a fit. Figure 1 draws the shifts against $k_{\rm max}$.

- $\Delta d_{\rm com} = 62$ Mpc: safe at 0.05 on every rung, for the calibrated truth and for every truth. The largest shift at that cut is 0.171σ (A_s, tied rung, $b_2 - 1$).
- $\Delta z = 0.02$: fails. At 0.05 the ultraviolet-fraction criterion keeps no band of any spectrum, so there is no fit; at 0.075 every rung already fails.
- $\Delta z = 0.05$: safe at 0.05 for the tied and baseline rungs under the calibrated truth only. The pessimistic and independent rungs fail there, with (w_0, w_a) shifts of 0.474 and 0.559, and every rung fails for some bracket.

- $\Delta z = 0.2$: fails everywhere, already at 0.05, with (w_0, w_a) shifts of 0.413 (tied) and 0.541 (the other rungs). Its groups hold one or two bins each, so the free amplitude and tilt already free every bin, and the baseline, pessimistic and independent rungs coincide.

Layout	tied	baseline	pessimistic	independent
$\Delta d_{\rm com} = 62$ Mpc	0.05 / 0.05	0.05 / 0.05	0.05 / 0.05	0.05 / 0.05
$\Delta z = 0.02$	fails / fails	fails / fails	fails / fails	fails / fails
$\Delta z = 0.05$	0.05 / fails	0.05 / fails	fails / fails	fails / fails
$\Delta z = 0.2$	fails / fails	fails / fails	fails / fails	fails / fails

Table 1: The largest safe $k_{\rm max}$ [h/Mpc] of a linear-bias fit to the one-loop truth: the largest scanned cut at which every shift, (w_0, w_a) in its ellipse and Ω_m and A_s in their own marginalized σ, stays below $\sigma/3$, at that cut and at every smaller cut with a fit. Each entry reads "calibrated truth / every truth". "fails": the smallest fitted cut already fails. On $\Delta z = 0.02$ the cut at 0.05 keeps no band, so the smallest fitted cut is 0.075. $k_{\rm max}$ quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{\rm max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

At the v1 cut, every rung is biased. At $k_{\rm max} = 0.075\,h$/Mpc, under the calibrated truth, every rung on every layout shifts some parameter by about 1σ or more, and no bracket brings any rung on any layout under the threshold. On $\Delta d_{\rm com} = 62$ Mpc the fit moves Ω_m by 1.98–2.42σ, (w_0, w_a) by 0.681–1.13 and A_s by 0.868–1.18 across the four rungs (Appendix B).

Why more bias freedom does not help. A linear bias multiplies a bin's power spectrum by the constant b_i^2 at every k. The loop terms and the counterterm $k^2 P_{\rm mm}$ of Eq. (2) have shapes different from $P_{\rm mm}$, so inside one bin the truth differs from every linear model by a scale dependence that no choice of b_i removes. The fit pushes that residual into the cosmological parameters, which do change the shape of the band powers. Freeing the biases changes which parameter takes the residual, not whether one does. On $\Delta d_{\rm com} = 62$ Mpc, going from the tied to the independent rung lowers the Ω_m shift from 2.42 to 1.98 and raises the (w_0, w_a) shift from 0.681 to 1.13. On $\Delta z = 0.05$ the baseline rung moves the error from Ω_m (0.162, against 2.32 for tied) to (w_0, w_a) (1.80, against 0.865).

The price of the safe cut. On $\Delta d_{\rm com} = 62$ Mpc the figure of merit ${\rm FoM}(w_0, w_a) = [\det {\rm Cov}(w_0, w_a)]^{-1/2}$ falls from 32.9, 18.2, 11.4 and 10.7 at 0.075 to 2.19, 1.54, 0.855 and 0.145 at 0.05 (tied, baseline, pessimistic, independent): 1–8% of its value, far below the LSST Y1 3x2pt reference of 17.26. The loss is part of why the cut at 0.05 passes: it keeps fewer bands, on larger scales, where the one-loop terms are smaller and the data less precise. The absolute FoMs carry the covariance mismatch of the v1 note (Appendix D).

The v1 accuracy test for Ω_m and A_s. Figure 2 repeats the v1 test, with linear truths at $k_{\rm max} = 0.075\,h$/Mpc: a known bias drift (the $r < 19.5$ flux limit in BGS, a conserved galaxy population in the other groups) plus a residual drawn from the baseline or the pessimistic prior, 2000 draws with the v1 random seed, each fitted with every rung. The verdict for Ω_m and A_s is the verdict for (w_0, w_a); the numbers are in the v1 note's appendix "The same test for Ω_m and A_s". The tied rung is excluded: the conserved-population drift alone moves Ω_m

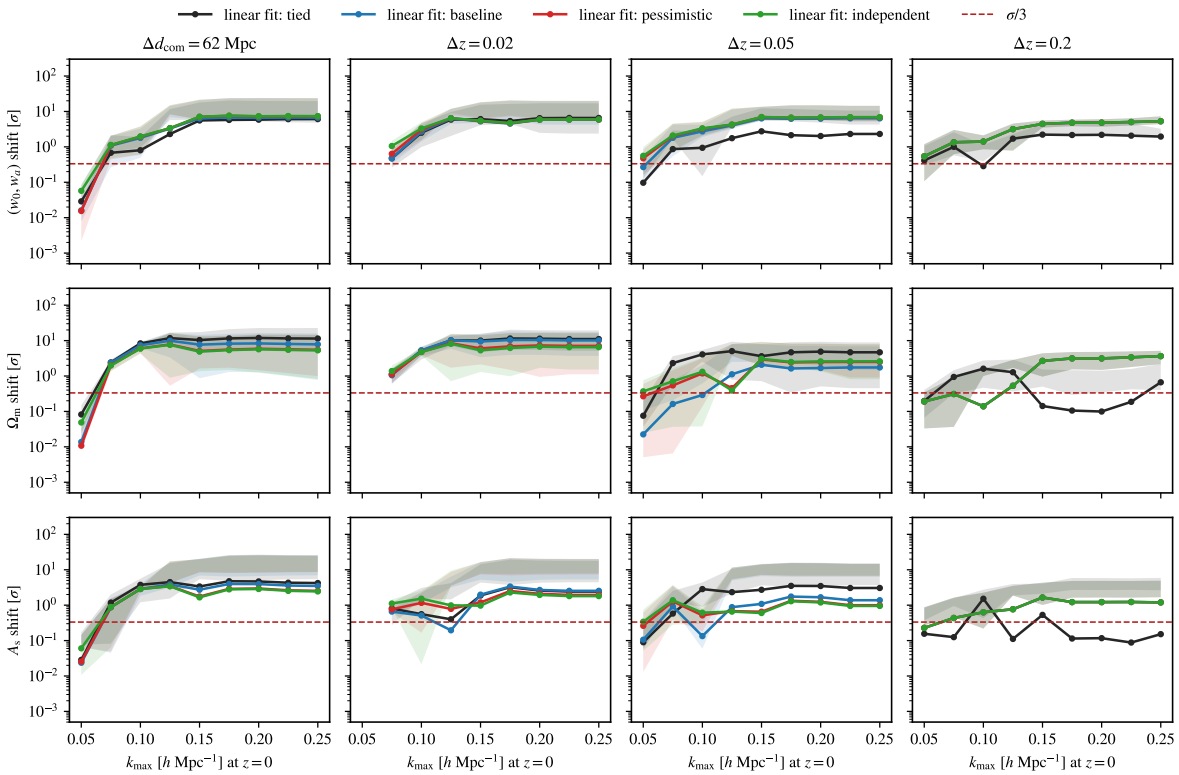


Figure 1: The shift of a linear-bias fit to the one-loop truth against the scale cut. Columns: the four bin layouts. Rows: the (w_0, w_a) shift in units of the rung's marginalized ellipse, and the Ω_m and A_s shifts in units of their own marginalized σ. Lines: the calibrated truth (b_2 on its Lazeyras calibration, b_{s^2} on co-evolution, no counterterm), one color per rung of the linear-bias ladder. Bands: the range spanned by the four brackets, every chunk's b_2 offset at ± 1 and $\alpha_0 = \pm 12.5\, h^{-2}\,\mathrm{Mpc}^2$. Dashed line: the $\sigma/3$ threshold. On $\Delta z = 0.02$ the cut at 0.05 keeps no band, so it has no point. On $\Delta z = 0.2$ the baseline, pessimistic and independent rungs coincide, and the independent line covers the other two. $k_{\max}$ quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{\max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

by 3.99, 5.21, 11.7 and 6.42σ, and A_s by 3.79, 9.90, 5.82 and 0.58σ, on $\Delta d_{\mathrm{com}} = 62$ Mpc, $\Delta z = 0.02$, 0.05 and 0.2. The error inflation is the factor by which a rung's quoted error would have to grow to cover the truth as often as it claims. It is 1 for the baseline rung under its own scatter, but under pessimistic scatter its Ω_m error bar is up to 2.75 times too small. The pessimistic rung is honest or conservative under both truths, with inflations of at most 1.04.

Recommendation.

- Linear bias is acceptable only at $k_{\max} \leq 0.05\, h/\mathrm{Mpc}$, and of the four layouts robustly, on every rung and for every truth, only on $\Delta d_{\mathrm{com}} = 62$ Mpc. On $\Delta z = 0.05$ it passes only for the tied and baseline rungs and only on the calibrations; on $\Delta z = 0.02$ and 0.2 no scanned cut is safe.
- That cut keeps less than a tenth of the FoM, so it is not a useful operating point.

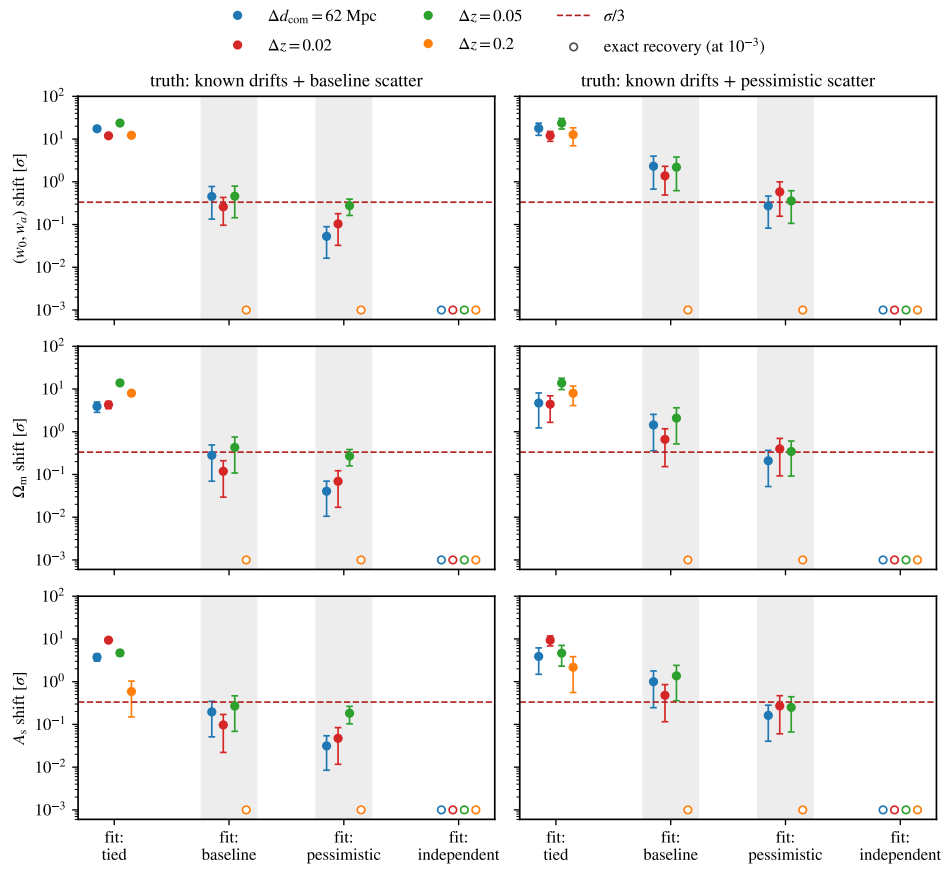


Figure 2: The v1 accuracy test scored for (w_0, w_a), Ω_m and A_s. The truths are linear: the true bias of bin i is the tied value times $1+\delta_i^{\rm drift}+\varepsilon_i$, with the drift the $r < 19.5$ flux-limit deviation in BGS and the conserved-population deviation in the other groups, and the residual ε_i drawn from the baseline prior (left column) or the pessimistic prior (right column); 2000 truths per column. Each truth is fitted with every rung (horizontal axis), every rung's bias prior centered on the tied shape. Rows: the (w_0, w_a) shift in units of the rung's ellipse, and the Ω_m and A_s shifts in units of their own marginalized σ. Points: the mean shift over the truths; bars: the 16th–84th percentiles; colors: the four bin layouts; dashed line: the $\sigma/3$ threshold. Gray columns mark the baseline and pessimistic rungs. In the left column the baseline rung, and in the right column the pessimistic rung, has a prior equal to the truth's scatter: there the mean shift is not a bias, because the rung's error already covers that scatter, and its error inflation is 1. Open markers on the floor at 10^{-3}: exact recovery, zero shift; the independent rung absorbs any such truth, and on $\Delta z = 0.2$ so do the baseline and pessimistic rungs. $k_{\rm max} = 0.075\,h/$Mpc quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{\rm max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

- At $k_{\rm max} = 0.075\,h/$Mpc the forecast should use the one-loop model A. The companion note `notes/correlated_bias_prior_v2`, being written, redoes the v1 ladder and accuracy test under model A.
- The v1 note's FoMs at $0.075\,h/$Mpc remain the linear-bias reference, not a final forecast.

A Method

A.1 The truths

Each truth is a full model A data vector at the project's fiducial cosmology. Model A is defined in the Recommendation section of `notes/higher_order_bias`: per-bin linear biases $b_{1,i}$, quadratic and tidal biases on calibrations of $b_{1,i}$ plus group offsets, a counterterm and a stochastic amplitude per group. The truths of this note set the linear biases to their fiducial values $b_{1,i} = B_g/D(\bar{z}_i)$ and choose the rest:

- **calibrated**: $b_{2,i}$ on the Lazeyras et al. separate-universe fit [3], $b_2(b_1) = 0.412 - 2.143\,b_1 + 0.929\,b_1^2 + 0.008\,b_1^3$; $b_{s^2,i} = -\frac{4}{7}(b_{1,i} - 1)$, the co-evolution value of a population formed with local bias and moved by gravity [4]; $b_3 = 0$; $\alpha_0 = 0$; no stochastic term;
- $b_2 + 1$ and $b_2 - 1$: the calibrated truth with every chunk's offset $\Delta_{2,g} = b_2 - b_2(b_1)$ set to $+1$ or -1, model A's prior width;
- $\alpha_0 = \pm 12.5\,h^{-2}\,\mathrm{Mpc}^2$: the calibrated truth with this counterterm in every chunk. It is the prior width of the DESI DR1 full-shape analysis [5]: $\alpha_0 k^2 = 0.5$ at $k = 0.2\,h/\mathrm{Mpc}$, a 50% correction to $P_{\rm lin}$ there [6].

The vectors come from CosmoLike's one-loop engine, at the same accuracy settings as the cached linear derivatives, and with the exact white-noise split of CosmoLike's low-rank engine. That engine factorizes $P_{gg} - P_\infty$, where P_∞ is the negative white-noise constant that the subtracted loop terms tend to at high k, and P_∞ reaches C_ℓ through Limber, which is exact for a constant. The subsection "The white-noise limit of the renormalized loops" of `notes/higher_order_bias` derives the constant and the treatment.

A.2 The fit: linear response of each rung

Let m be the expected data vector, C its covariance on the kept entries (its inverse rescaled from 14,000 to 12,300 square degrees), and $\mathsf{D} = \partial m/\partial p$ the matrix of derivatives of the linear-bias model with respect to the fitted parameters p. These are the seven cosmological parameters ($10^9 A_s$, n_s, H_0, Ω_b, Ω_m, w_0, w_a, flat priors), the rung's bias parameters, and the per-group velocity dispersions with their Gaussian priors. The rung's Fisher matrix is $F = \mathsf{D}^\mathsf{T}\mathsf{C}^{-1}\mathsf{D} + F_{\rm priors}$, where $F_{\rm priors}$ holds the velocity-dispersion priors and, on the baseline and pessimistic rungs, the projected precision of the bias prior derived in the v1 note.

If the data are the truth $m_{\rm truth}$ while the model at the fiducial predicts $m_{\rm lin}$, then to first order in the difference the best fit moves by

$$\delta p \;=\; F^{-1}\,\mathsf{D}^\mathsf{T}\,\mathsf{C}^{-1}\big(m_{\rm truth} - m_{\rm lin}\big). \tag{3}$$

The derivation is one line: at the old best fit the gradient of the log posterior becomes $\mathsf{D}^\mathsf{T}\mathsf{C}^{-1}(m_{\rm truth} - m_{\rm lin})$ instead of zero, and F is the curvature that converts that gradient into a parameter displacement. The tied rung uses the cached tied derivatives, one bias column per group. The other rungs use the per-bin derivatives, one bias column per bin. Both sets come from the linear notebook (`EXAMPLE_EVALUATE_FISHER.ipynb`).

A.3 The scores and the threshold

With δw the (w_0, w_a) entries of δp and C_w the (w_0, w_a) block of F^{-1}, the rung's marginalized 2×2 covariance, the (w_0, w_a) shift is

$$\Delta_w \;=\; \sqrt{\delta w^{\mathsf{T}}\, \mathsf{C}_w^{-1}\, \delta w}, \tag{4}$$

the displacement of the dark-energy ellipse in units of its own size. For a single parameter p the shift is $\Delta_p = |\delta p|/\sigma_p$, with $\sigma_p = \sqrt{(F^{-1})_{pp}}$ its marginalized error. The threshold is $1/3$, as in the v1 note. A shift of $\sigma/3$ added in quadrature to a 1σ error gives $\sqrt{1 + 1/9} = 1.05$, so below the threshold the total error grows by at most about 5%. Both the response and the error use the same rung's F.

A.4 The cut, cuts without data, and the safe $k_{\max}$

The kept entries at each $k_{\max}$ are the shipped mask intersected with the ultraviolet-fraction cut of Eq. (1), and the inverse covariance is rebuilt on them. The shipped mask caps every bin at $0.1185\,\mathrm{Mpc}^{-1}$, about $0.176\,h/\mathrm{Mpc}$ at the fiducial $h = 0.6732$, with no growth scaling. At the largest cuts the shipped mask, not the ultraviolet-fraction criterion, sets more and more of the kept bands, and the curves of Figure 1 flatten.

At small $k_{\max}$ some bins lose every band. Their bias columns then hold only finite-difference noise, and keeping them would make F numerically singular. A parameter whose information at a cut, data and prior, is below 10^{-10} of its information on the full shipped mask therefore leaves the fit at that cut. A cut that leaves no data for the cosmological parameters gives no fit. This happens on $\Delta z = 0.02$ at $k_{\max} = 0.05\,h/\mathrm{Mpc}$, where the ultraviolet-fraction criterion keeps no band of any spectrum.

The largest safe $k_{\max}$ of Table 1 is the largest scanned cut at which every score of every truth considered is below $1/3$, at that cut and at every smaller cut with a fit. A cut without a fit neither passes nor fails, so the scan skips it.

B The shift tables

Table 2 lists the shifts of a linear fit to the calibrated truth at the two smallest cuts, with each rung's FoM. Table 3 lists, at $k_{\max} = 0.05\,h/\mathrm{Mpc}$, the largest shift over all five truths and the three scores, which decides the "every truth" column of Table 1.

Four readings of the tables.

The 62 *Mpc pass is comfortable.* At $k_{\max} = 0.05\,h/\mathrm{Mpc}$ no truth brings any rung above 0.171, half the threshold. The step to 0.075 multiplies the shifts of the calibrated truth by factors of 15 to 190.

The $\Delta z = 0.05$ *pass is fragile.* The tied and baseline rungs pass on the calibrations, the baseline with a (w_0, w_a) shift of 0.268, and both fail for one bracket: the tied rung on A_s at $\alpha_0 = -12.5$, the baseline on (w_0, w_a) at $b_2 - 1$. The pessimistic and independent rungs fail on the calibrations already.

More bias freedom moves the error. At 0.075 on $\Delta z = 0.05$ the tied rung puts most of the residual into Ω_m (2.32) and the per-bin rungs into (w_0, w_a) (1.80–2.09), while their Ω_m shift drops to 0.162–0.710. On $\Delta z = 0.2$ the tied rung has the smaller A_s shift (0.125 against 0.435) but no rung passes.

		$k_{\max} = 0.05\,h/\mathrm{Mpc}$				$k_{\max} = 0.075\,h/\mathrm{Mpc}$			
Layout	Rung	Δ_w	Ω_m	A_s	FoM	Δ_w	Ω_m	A_s	FoM
$\Delta d_{\mathrm{com}} = 62$ Mpc	tied	0.0289	0.0821	0.0285	2.19	0.681	2.42	1.18	32.9
	baseline	0.0161	0.0137	0.0237	1.54	1.08	2.35	0.868	18.2
	pessimistic	0.0154	0.0108	0.0253	0.855	1.12	2.03	0.884	11.4
	independent	0.0573	0.0488	0.0604	0.145	1.13	1.98	0.887	10.7
$\Delta z = 0.02$	tied		no band kept			0.474	1.12	0.806	19.9
	baseline		no band kept			0.477	1.05	0.659	12.5
	pessimistic		no band kept			0.636	1.12	0.762	7.41
	independent		no band kept			1.06	1.38	1.11	4.02
$\Delta z = 0.05$	tied	0.0971	0.0756	0.0898	1.90	0.865	2.32	0.579	8.29
	baseline	0.268	0.0225	0.106	0.655	1.80	0.162	0.920	1.96
	pessimistic	0.474	0.267	0.261	0.374	2.02	0.542	1.28	1.33
	independent	0.559	0.369	0.340	0.317	2.09	0.710	1.38	1.23
$\Delta z = 0.2$	tied	0.413	0.198	0.156	0.733	1.00	0.941	0.125	1.91
	baseline	0.541	0.190	0.228	0.205	1.35	0.310	0.435	0.515
	pessimistic	0.541	0.190	0.228	0.205	1.35	0.310	0.435	0.515
	independent	0.541	0.190	0.228	0.205	1.35	0.310	0.435	0.515

Table 2: The shifts of a linear-bias fit to the calibrated one-loop truth: Δ_w of Eq. (4) in units of the rung's (w_0, w_a) ellipse, and the Ω_m and A_s shifts in units of their own marginalized σ; the threshold is 1/3. FoM: FoM(w_0, w_a) of the rung at that cut. "no band kept": the cut leaves no data, so there is no fit. On $\Delta z = 0.2$ the three lower rungs coincide. $k_{\max}$ quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{\max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

At the safe cut, the brackets below the calibration decide. The negative brackets, $b_2 - 1$ and $\alpha_0 = -12.5$, give the largest shift in every fitted case of Table 3. At 0.075 every truth fails on every rung and layout; there the brackets reach 3.63σ in Ω_m (tied rung, $b_2 + 1$, $\Delta z = 0.05$) and 4.55 in (w_0, w_a) (independent rung, $b_2 - 1$, $\Delta z = 0.05$).

C Numerical checks

- **One fiducial vector.** The linear vector computed in the same session as the truths equals the cached reference of the linear notebook exactly on $\Delta d_{\mathrm{com}} = 62$ Mpc, $\Delta z = 0.02$ and 0.05. On $\Delta z = 0.2$ the largest relative difference is 4.5×10^{-3}, at a cross spectrum whose band powers are nearly zero; the median relative difference is 2×10^{-5}, and the difference is $\Delta\chi^2 = 6 \times 10^{-5}$ on the shipped mask. The cached derivatives and the truths therefore share one fiducial.

- **Size of the one-loop sector.** Over the data vector, the one-loop part of the calibrated truth is a median 1.9%, 1.6%, 2.3% and 1.9% of the linear band powers on $\Delta d_{\mathrm{com}} = 62$ Mpc, $\Delta z = 0.02$, 0.05 and 0.2.

- **The rungs are the v1 rungs.** The FoMs at $k_{\max} = 0.075\,h/\mathrm{Mpc}$ in Table 2 equal the v1 ladder, from 32.9, 18.2, 11.4 and 10.7 on $\Delta d_{\mathrm{com}} = 62$ Mpc to 1.91 and 0.515 on $\Delta z = 0.2$.

- **The truths are the v1 truths.** The drifts and the 2000 random truths are drawn

Layout	Rung	largest shift	score	truth
$\Delta d_{\rm com} = 62$ Mpc	tied	0.171	A_s	$b_2 - 1$
	baseline	0.147	A_s	$b_2 - 1$
	pessimistic	0.142	A_s	$b_2 - 1$
	independent	0.144	A_s	$\alpha_0 = -12.5$
$\Delta z = 0.05$	tied	0.345	A_s	$\alpha_0 = -12.5$
	baseline	0.492	(w_0, w_a)	$b_2 - 1$
	pessimistic	0.838	(w_0, w_a)	$b_2 - 1$
	independent	0.982	(w_0, w_a)	$b_2 - 1$
$\Delta z = 0.2$	tied	1.06	(w_0, w_a)	$b_2 - 1$
	baseline	1.19	(w_0, w_a)	$\alpha_0 = -12.5$
	pessimistic	1.19	(w_0, w_a)	$\alpha_0 = -12.5$
	independent	1.19	(w_0, w_a)	$\alpha_0 = -12.5$

Table 3: The largest shift at $k_{\rm max} = 0.05\,h/\text{Mpc}$ over the five truths and the three scores, with the score and the truth that give it; α_0 in $h^{-2}\,\text{Mpc}^2$. $\Delta z = 0.02$ has no fit at this cut. $k_{\rm max}$ quoted at $z = 0$; each bin is cut by the UV-fraction criterion at $k_{\rm max}/D(\bar{z}_i)$ ($\alpha = 0.05$) within the shipped mask.

with the v1 seed and order. The matched cases reproduce the v1 (w_0, w_a) error inflation within 0.019 of 1, and the drift-only (w_0, w_a) shifts of the tied rung, 17.4, 12.4, 21.5 and 10.9 for the conserved-population drift, equal the v1 table.

- **The white-noise constant.** The truths use the exact projection of the one-loop white-noise constant (Appendix A.1). The higher-order note reports that this projection leaves linear-bias vectors bitwise unchanged, and the v1 note's 62 Mpc numbers at 0.075, computed before it, are reproduced here.

D Limitations

- **Linear response.** The shifts come from the first-order response of Eq. (3), not from a nonlinear refit. The verdicts rest on shifts either well below the threshold (62 Mpc at 0.05) or several times above it (0.075), where a second-order correction would have to be as large as the shift itself to change them. The near-threshold cases, on $\Delta z = 0.05$ and 0.2 at 0.05, are the ones a refit could move.

- **Theory truths, not mocks.** The truths are model A at and around its calibrations. The project is a theory paper and uses no mock catalogs, so a real sample whose higher-order biases sit outside the brackets is not covered.

- **No stochastic term.** The truths carry no non-Poisson shot noise and set $b_3 = 0$. A constant noise power projects to an ℓ-independent C_ℓ, which a linear bias cannot mimic either; its effect on the shifts is not measured here.

- **Real-space loops.** The truths add only the real-space one-loop terms of model A; the redshift-space one-loop terms of model B are absent from truth and fit alike.

- **Negative power in one bracket.** For BGS-like b_1 the $\alpha_0 = -12.5\,h^{-2}\,\mathrm{Mpc}^2$ truth makes P_{gg} negative above $k \approx 0.37\,h/\mathrm{Mpc}$. The full Limber term carries that region exactly, while the low-rank non-Limber correction floors it. This is immaterial at $k_{\max} \leq 0.075\,h/\mathrm{Mpc}$, where the conclusions are drawn.

- **The covariance.** As in the v1 note, the covariance is built with each group's raw bias while the data vector uses $b_i = B_g/D(\bar{z}_i)$, which inflates the absolute FoMs by about 2.4–3.2. The shifts and FoMs here use that covariance and will change when it is rebuilt.

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